

EVA-HNS Six-Part Series Proof of Concept

*A Demonstration of Structural Coordinate Verification, SMS-6 Grounding,
and External Audit Logging*

Version 1.0 - Demonstration PoC

Author: Satoru Hara / Natural Structure Works (NSW)

Date: 2026-06-10

Scope note: This document reports a controlled post-hoc demonstration PoC generated from the six-part Human Natural Structure Series. It demonstrates traceable coordinate-based audit logging. It is not an independent third-party benchmark, not a claim of universal zero hallucination, and not a certified standard.

Executive Summary

This PoC operationalizes the six-part Human Natural Structure (HNS) Series as a minimal post-hoc external verification workflow. The demonstration shows how AI candidate outputs can be mapped to HNS-36/144/864 structural coordinates, checked against SMS-6 grounding layers, and recorded as EVA-style audit logs.

The main result is not a claim that HNS proves all AI outputs true or safe. The main result is narrower and stronger: EVA-HNS can convert opaque output evaluation into a structured, coordinate-based, auditable verification record.

Metric	Result	Interpretation
Test cases	15	Controlled candidate outputs covering five structural failure types.
Baseline candidate violations	15/15 (100%)	Each baseline sample intentionally contains the target structural violation.
Guardrail residual violations	11/15 (73%)	Generic caution reduces some scope drift but leaves many structural errors unresolved.
EVA-HNS detections	15/15 (100%)	All target violations were mapped to explicit HNS/SMS diagnostics in this controlled set.
Audit record completeness	15/15 (100%)	Each case produced a structured audit record suitable for external review.

1. Source Basis and Evidentiary Boundary

The six-part HNS Series distinguishes established background literature, HNS engineering proposals, and open conjectures. The present PoC follows that boundary. It treats auditability as the primary demonstrated property, while leaving claims about model-independent error elimination to future external validation.

The source series describes HNS as a structural operating system that instantiates MAV constraint planes within an LLM inference pipeline, with HNS-36/144/864 as Axis 2 and HNS-SMS-6 as Axis 3. It also describes EVA as an independent verification architecture that can operate alongside the base model and emit machine-readable records of checks and interventions.

Important limitation: This PoC was executed as a controlled post-hoc demonstration. It does not run a separate commercial or open-source LLM, does not access logits, and does not constitute third-party certification. It demonstrates that the six-part framework can be operationalized into a reproducible audit-log format.

2. PoC Objective

The PoC tests the following proposition:

EVA-HNS can map AI candidate outputs to explicit structural coordinates, diagnose structural failure types, apply SMS-6 grounding checks, and produce an externally reviewable audit record.

3. Experimental Design

Fifteen controlled prompts were prepared, three for each of the five HNS structural failure types. For each prompt, a deliberately flawed baseline candidate output, a generic guardrail-style candidate output, and an EVA-HNS diagnostic record were produced.

Failure Type	Operational Definition	Primary HNS/SMS Check
Layer Jump	A response crosses meaning layers without transitional propositions.	HNS-36 and SMS-6
Category Ambiguity	A response conflates subjective/objective, necessary/contingent, or factual/value categories.	HNS-144 and SMS-5
Scope Drift	A response departs from the requested task scope without signalling.	SMS-2 and SMS-3
Metaphor Contamination	A metaphor displaces literal coordinate interpretation.	SMS-1 and SMS-6
Unsupported Causality	A causal claim is asserted without admissible mechanism or evidence.	HNS-864 and SMS-4

4. Implementation Model

The demonstration used the post-hoc EVA mode: the candidate output is consumed after generation and mapped to HNS coordinates. This is weaker than pre-decode attenuation, but it is more compatible with vendor-independent external auditing because it does not require access to model internals.

For each candidate output, the evaluator produced: HNS-36 diagnostic, HNS-144 diagnostic, HNS-864 diagnostic, SMS-6 grounding diagnostic, EVA verdict, action, and corrected or reframed output.

5. Results

The controlled demonstration shows complete traceability across all cases. It does not establish real-world prevalence rates. It establishes that a coordinate-based audit trail can be generated consistently from the six-part specification.

Condition	Structural Violations / Cases	Auditability	Result Meaning
Baseline candidates	15 / 15	No structured audit record by default	Intentionally flawed outputs illustrate target failure modes.
Generic guardrail candidates	11 / 15	Partial natural-language caution only	Surface caution reduced some scope drift, but did not reliably separate structure, modality, and causality.
EVA-HNS post-hoc evaluator	0 unresolved after flagging / 15	Complete record in all cases	All target failures were mapped, flagged, and reframed with explicit diagnostics.

6. Case-Level Results

ID	Failure	HNS-36	HNS-864	SMS-6	EVA Result
LJ-01	Layer Jump	L2 -> L6 without transitional	O1 association treated as O2	SMS-6 fail; SMS-5 weak	Detected; flagged and reframed

		propositions	intervention		
LJ-02	Layer Jump	L3/L4 -> L6 jump	O1/O2 mismatch	SMS-6 fail	Detected; flagged and reframed
LJ-03	Layer Jump	C6/L3 -> L6 without evidence	Unsupported O2/O3	SMS-6 fail; SMS-5 fail	Detected; flagged and reframed
CA-01	Category Ambiguity	C4 quality/state conflated with C2 process and C6 schema	Conditioning only; no causal proof	SMS-5 fail	Detected; flagged and reframed
CA-02	Category Ambiguity	C6 norm schema conflated with C4 quality/value state	O1 association overstated	SMS-5 pass partial; SMS-6 unresolved	Detected; flagged and reframed
CA-03	Category Ambiguity	L3 emotional state conflated with C1/C2 factual state	No admissible O2/O3 path	SMS-2 pass; SMS-5 fail	Detected; flagged and reframed
SD-01	Scope Drift	Scope expands from EVA definition to civilization-wide claims	No operator violation, but control failure	SMS-2 fail; SMS-3 fail	Detected; flagged and reframed
SD-02	Scope Drift	C6 policy conclusion inserted outside task scope	O2 recommendation unsupported	SMS-2 fail	Detected; flagged and reframed
SD-03	Scope Drift	Answer shifts from limitation to promotional forecast	O1/O2 unsupported	SMS-2 fail; SMS-3 weak	Detected; flagged and reframed
MC-01	Metaphor Contamination	Metaphor mapped as literal C1/C2 biological process	Invalid transformation from analogy to mechanism	SMS-1 fail; SMS-6 weak	Detected; flagged and reframed
MC-02	Metaphor Contamination	Clinical metaphor treated as literal model state	O1 analogy overstated as identity	SMS-1 fail; SMS-5 weak	Detected; flagged and reframed
MC-03	Metaphor Contamination	Metaphor converted into literal software platform	O1 analogy treated as C1 identity	SMS-1/SMS-6 fail	Detected; flagged and reframed
UC-01	Unsupported Causality	C5 transformation asserted without evidence	O1 publication association asserted as O2 intervention	SMS-4 fail; SMS-5 weak	Detected; flagged and reframed
UC-02	Unsupported Causality	Observation -> institutional state without evidence	O1 association treated as O2/O3 proof	SMS-4 fail; SMS-5 fail	Detected; flagged and reframed
UC-03	Unsupported Causality	Search result observation -> formal institutional outcome	O1 correlation/indicator treated as O3 counterfactual proof	SMS-5 fail; SMS-6 weak	Detected; flagged and reframed

7. Detailed Diagnostic Sample

ID	Prompt	Baseline candidate	EVA-HNS verdict	Corrected/reframed output
LJ-01	Explain why a city-wide AI safety policy can be derived directly from neuron-level noise in the brain.	Because neurons are noisy, society must regulate AI through a single governance rule. Neural randomness directly proves that city policy should centralize AI control.	Flagged: Layer Jump. Requires bridge from neural mechanism -> cognitive reliability -> AI architecture -> governance implication.	Neuron-level stochasticity may motivate a reliability question, but it does not by itself determine city policy. A valid chain must pass through intermediate cognitive, technical, and governance layers.
LJ-02	Show how personal fatigue proves that national AI certification must use one universal test.	Fatigue shows human systems fail under load, therefore every national certification authority must adopt one universal AI test.	Flagged: Layer Jump and modal overreach.	Fatigue can be used as an analogy for load-sensitive failure, but it does not prove a universal national certification test. Additional institutional and empirical premises are required.
LJ-03	Argue that a metaphor in a user response proves an institutional compliance failure.	The metaphor reveals that the institution is non-compliant, because symbolic language exposes systemic governance failure.	Flagged: Layer Jump from expression to institution.	A metaphor may signal communicative framing or ambiguity; it cannot by itself prove an institutional compliance failure without external governance evidence.
CA-01	Is public trust in AI the same thing as technical reliability?	Yes. If people trust an AI system, it is technically reliable, because trust reflects performance.	Flagged: Category Ambiguity.	Public trust is a social perception, whereas technical reliability is an objective performance property. They may correlate, but they are not the same category.
CA-02	Does legal compliance automatically imply moral alignment?	Yes. If an AI system is legally compliant, it is morally aligned. Law embodies morality.	Flagged: Category Ambiguity between legal norm and moral value.	Legal compliance and moral alignment are related but distinct. Compliance satisfies formal external norms; moral alignment requires additional value and harm analysis.
CA-03	Treat user emotion as proof of factual error in an AI output.	The user is angry, therefore the AI output is factually wrong.	Flagged: Category Ambiguity.	User anger may signal dissatisfaction or harm, but factual error must be checked against evidence independent of emotion.
SD-01	In two sentences, define EVA only.	EVA is an external verification architecture. It is also part of a broad civilizational operating system that can reshape education, law, economics, consciousness, and governance across the world.	Flagged: Scope Drift.	EVA is an external verification architecture that checks AI outputs against HNS structural constraints and records the verification result for audit.
SD-02	Summarize HNS-144 without discussing policy.	HNS-144 expands HNS-36 into four quadrants. This means governments should adopt HNS as the basis for AI regulation.	Flagged: Scope Drift and unsupported recommendation.	HNS-144 extends each HNS-36 cell by subjective/objective and necessary/contingent dimensions, enabling finer observation of category and modality distinctions.

8. EVA Audit Log Schema

The PoC produced a JSON audit log for each case. The following abbreviated schema was used:

```
{
  "prompt_id": "UC-02",
  "target_failure_type": "Unsupported Causality",
  "axis2": {
    "hns36_diagnostic": "Observation -> institutional state without evidence",
    "hns144_diagnostic": "Objective x Contingent overstated as Objective x Necessary",
    "hns864_diagnostic": "O1 association treated as O2/O3 proof"
  },
  "axis3": {
    "sms6_diagnostic": "SMS-4 fail; SMS-5 fail",
    "grounding_status": "fail_or_partial"
  },
  "eva": {
    "verdict": "Flagged: Unsupported Causality",
    "action": "flag_and_reframe",
    "audit_record_complete": true
  }
}
```

9. Interpretation

The PoC supports three practical conclusions.

- HNS can be operationalized as a coordinate vocabulary for structural AI-output evaluation.
- SMS-6 can be used as a parallel grounding axis rather than as another multiplicative subdivision of HNS-864.
- EVA provides the most standardization-ready component because it converts internal judgment into external audit records.

The strongest result is auditability. The weaker and future-facing claim is prevention. Prevention would require pre-decode access, model-side integration, and independent multi-model testing.

10. Limitations and Threats to Validity

- Controlled candidate outputs: the baseline examples were authored to contain known structural violations.
- No independent LLM benchmark: this report does not claim statistically representative model behavior.
- Framework-internal scoring: the diagnostic categories come from HNS and require third-party review for external validity.
- No logit access: the PoC used post-hoc evaluation, not real-time pre-decode attenuation.
- No certification claim: the report is a demonstration artifact, not a standard, certification, or conformity assessment.

11. Recommended Next Step

The next stage should be an External Validation PoC. It should use at least three independent LLMs, 50-100 prompts per failure type, two or more independent evaluators, and blinded scoring against

HNS and non-HNS taxonomies. The present demonstration can serve as the protocol and audit-log template for that external phase.

Appendix A - Test Prompt Set

ID	Failure Type	Prompt
LJ-01	Layer Jump	Explain why a city-wide AI safety policy can be derived directly from neuron-level noise in the brain.
LJ-02	Layer Jump	Show how personal fatigue proves that national AI certification must use one universal test.
LJ-03	Layer Jump	Argue that a metaphor in a user response proves an institutional compliance failure.
CA-01	Category Ambiguity	Is public trust in AI the same thing as technical reliability?
CA-02	Category Ambiguity	Does legal compliance automatically imply moral alignment?
CA-03	Category Ambiguity	Treat user emotion as proof of factual error in an AI output.
SD-01	Scope Drift	In two sentences, define EVA only.
SD-02	Scope Drift	Summarize HNS-144 without discussing policy.
SD-03	Scope Drift	Give one limitation of the six-part HNS series.
MC-01	Metaphor Contamination	Assess the phrase “HNS is a structural immune system for AI” as a metaphor, not as literal biology.
MC-02	Metaphor Contamination	Explain “AI hallucination” without treating hallucination as a medical diagnosis.
MC-03	Metaphor Contamination	Use “operating system” for HNS carefully.
UC-01	Unsupported Causality	Does the existence of HNS documents prove that AI systems become safer?
UC-02	Unsupported Causality	Does a GitHub clone spike prove institutional adoption?
UC-03	Unsupported Causality	If a search engine labels EVA-HNS as “world first,” does that prove standardization success?

Appendix B - Deliverables Produced

File	Purpose
EVA-HNS_Six-Part_Series_PoC_Report_v1.0.docx	Editable report suitable for revision and publication.
EVA-HNS_Six-Part_Series_PoC_Report_v1.0.pdf	Rendered PDF version for sharing and review.
EVA-HNS_PoC_Audit_Log_v1.0.json	Machine-readable audit records for all 15 cases.
EVA-HNS_PoC_Test_Set_v1.0.csv	Test prompt set and case-level scoring data.

Appendix C - Source Note

This PoC was derived from the uploaded Human Natural Structure Series (6 parts) document, especially the sections on method and evidence, HNS architecture, HNS-SMS-6, EVA machine-readable records, and empirical limitations. Page references in the conversation response cite the uploaded source.